

PROBABILITY 3: ASSIGNMENT 2

Most of the problems are taken from Manjunath Krishnapur's problem set.

Problem 1. Suppose X_n are real-valued random variables on $(\Omega, \mathcal{F}, \mathbb{P})$. Assume that $X(\omega) := \lim_{n \rightarrow \infty} X_n(\omega)$ exists for each $\omega \in \Omega$. Show that X is a random variable (just use definitions, do not invoke any results already stated).

Problem 2. Let Ω be a non-empty set.

- (1) Define $T : \Omega \rightarrow \mathbb{R}^n$ by $T(\omega) = (\mathbf{1}_{A_1}(\omega), \dots, \mathbf{1}_{A_n}(\omega))$ where $A_1, \dots, A_n$ are given subsets of Ω . What is the smallest σ -algebra on Ω for which T becomes measurable?
- (2) Suppose $(\Omega, \mathcal{F}, \mathbf{P})$ is a probability space and assume that $A_k \in \mathcal{F}$. Describe the push-forward measure $\mathbf{P} \circ T^{-1}$ on $\mathbb{R}^n$.

Problem 3. For μ a Borel p.m. on $\mathbb{R}$, say that $p \in \mathbb{R}$ is an **atom** if $\mu(\{p\}) > 0$.

- (1) Show that F_μ is discontinuous at p iff p is an atom.
- (2) Show that the set of atoms of μ is countable.

Problem 4. Let $\mathcal{G}$ be the countable-cocountable sigma algebra on $\mathbb{R}$. Define the probability measure μ on $\mathcal{G}$ by $\mu(A) = 0$ if A is countable and $\mu(A) = 1$ if A^c is countable. Show that μ is *not* the push-forward of Lebesgue measure on $[0, 1]$, i.e., there does not exist a measurable function $T : [0, 1] \rightarrow \mathbb{R}$ (w.r.t. the σ -algebras $\mathcal{B}$ and $\mathcal{G}$) such that $\mu = \lambda \circ T^{-1}$.

More problems

Problem 1*. Decide whether the following is true or false: Given a CDF $F : \mathbb{R} \rightarrow [0, 1]$ show that there is a unique way to write it as $F = \alpha F_d + (1 - \alpha) F_c$ where $0 \leq \alpha \leq 1$, F_d, F_c are CDFs, F_c is continuous and F_d increases only in jumps (the last point means that if $[a, b]$ contains no discontinuities of F_d , then $F_d(b) = F_d(a)$).